

BAHRAM RAHMATI

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[Google Scholar](#)

EDUCATION

Oct 2020 – Sep 2023

Master of Science in Mechanical Engineering (Energy Conversion)

Semnan University, Iran

GPA: 19.2/20.0 – Ranked 1st in class

Thesis: “Experimental and Analytical Study on Performance Enhancement of Indirect Evaporative Cooler (IEC) Heat Exchanger for Conditions in Semnan”

Feb 2013 – Jul 2017

Bachelor of Science in Mechanical Engineering

Semnan University, Iran

Thesis: “CFD Study on Thermal Comfort by Simulation of an Under-Floor Air Distribution System in Airpak Software”

ACADEMIC EXPERIENCE

May 2024 – Present

Research Assistant

Semnan University, Iran

- Creation of an IEC setup at Semnan University’s Thermodynamic Laboratory
- Mentorship of students in measurement and tests with the IEC setup
- Calibration of temperature and humidity sensors to ensure accurate and reproducible results (maintenance)

Oct 2022 – Feb 2023

Teaching Assistant

Courses (Graduate level): Convection Heat Transfer | Research Methodology

Semnan University, Iran

- Delivered supplementary lectures of weekly course materials
- Provided tailored support, helping students grasp challenging concepts
- Defined and supervised individual final projects, aligning them with course objectives and real-world applications

SKILLS

- CFD & Numerical Simulations: ANSYS FLUENT | Airpak
- Programming Languages: C++ | Engineering Equation Solver (EES)
- Visualization & Graphical Works: Tecplot
- Industrial Drafting: SolidWorks
- Other Software: TRANSYS

LANGUAGES

- Persian (Native)
- English (IELTS 7.0): Speaking – 7.0 | Listening – 7.5 | Reading – 6.5 | Writing – 6.5

AWARDS

- Honored for outstanding academic and research contributions at the “Gratitude Ritual 2022” by Semnan University

JOURNAL ARTICLES

	Cited by	Year
1. Rahmati, B. , Heidarian, A. & Jadidi, A. M. "Investigation in performance of a hybrid under-floor air distribution with improved desk displacement ventilation system in a small office." <i>Applied Thermal Engineering</i> (Q1-ranked Impact Factor = 6.9) DOI: https://doi.org/10.1016/j.applthermaleng.2018.03.015	50	2018
2. Rahmati, B. , Jadidi, A. M. & Valipour, M. S. "Enhancing the performance of an M-cycle based tubular indirect evaporative cooler by mesh screens." <i>International Communications in Heat and Mass Transfer</i> (Q1-ranked Impact Factor = 6.4) DOI: https://doi.org/10.1016/j.icheatmasstransfer.2024.107575	7	2024
3. Rahmati, B. , Heidarian, A. & Jadidi, A. M. "The relation between airflow pattern and indoor air quality in a hybrid personalized ventilation with under-floor air distribution system." <i>Experimental Techniques</i> (Q2-ranked Impact Factor = 1.9) DOI: https://doi.org/10.1007/s40799-022-00595-0	5	2023
4. Arzani, S., Rahmati, B. & Jadidi, A. M. "Chilled ceiling effects on the indoor air quality in a room equipped with displacement ventilation system." <i>Journal of Heat and Mass Transfer Research</i> (Q2-ranked) DOI: https://doi.org/10.22075/jhmtr.2021.22921.1337	2	2021
5. Jadidi, A. M., Rahmati, B. & Heidarian, A. "Effect of opening diffuser and return vent location on air quality, thermal comfort and energy saving in desk displacement ventilation (DDV) system." <i>Journal of Heat and Mass Transfer Research</i> (Q2-ranked) DOI: https://doi.org/10.22075/jhmtr.2018.12784.1193	1	2018

CONFERENCES

1. Rahmati, B. (the presenter) , Heidarian, A. & Jadidi, A. M. "Investigation in performance of a hybrid personalized ventilation (PV) with under-floor air distribution (UFAD) system." <i>26th Annual International Conference of Iranian Society of Mechanical Engineers (ISME 2018)</i> Oral Presentation in English at Semnan University, Semnan, Iran	1	2018
2. Rahmati, B. (the presenter) , Heidarian, A. & Jadidi, A. M. "Optimization of the distance between the opening diffuser and return vent in a displacement ventilation system." (in Persian). <i>25th Annual International Conference of Iranian Society of Mechanical Engineers (ISME 2017)</i> Oral Presentation in Persian at Tarbiat Modares University, Tehran, Iran	---	2017

REFERENCES

Dr. Amir Mohammad Jadidi

Assistant Professor, Mechanical Engineering Department, Semnan University, Iran

Role: Main supervisor of BSc and MSc theses

Email: am.jadidi@semnan.ac.ir

Dr. Mohammad Sadegh Valipour

Professor, Dean of the Faculty of Mechanical Engineering, Semnan University, Iran

Role: Co-supervisor of MSc thesis and the supervisor for teaching assistantship

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